# Project - High Level Design

# On

# Deploy Manufacturing App to Kubernetes

Course Name: Devops

***Institution Name:*** Medicaps University – Datagami Skill Based Course

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# 1. Introduction

This document presents the system design for the **Deploy Node.js Retail App to Kubernetes** project, developed as part of the Datagami Skill Based Course at Medicaps University.

The project demonstrates a complete DevOps implementation involving containerization, cloud-native orchestration, and automated deployment of a **Node.js Retail Application** using AWS EKS.

The DevOps infrastructure ensures the application is reliably packaged, deployed, and updated on a production-grade AWS EKS Kubernetes cluster with **zero downtime on every release**.

## 1.1 Scope of the Document This document covers:

- System architecture and component interactions
- Docker containerization strategy for Node.js app
- Kubernetes deployment and service configuration
- CI/CD pipeline using GitHub Actions
- AWS EKS infrastructure setup
- Secrets and environment variable handling
- Security and design decisions
- API catalogue (health + retail endpoints)

## 1.2 Intended Audience

| Audience | Purpose |
|---|---|
| DevOps Engineers | Understand CI/CD pipeline & infra |
| Backend Developers | Understand containerization |
| Cloud Architects | Review AWS EKS design |
| Evaluators | Assess implementation |

| | |
|---|---|
| Maintainers | Extend system |

| | |
|---|---|
| **Project Mentors** | Review the overall system design for guidance and feedback |
| **Future Maintainers** | Onboard quickly and extend the infrastructure |

A basic understanding of Docker, Kubernetes, GitHub Actions, and AWS services is recommended for reading this document.

### 1.3 System Overview

The system can follow:

#### Option 1: Single Repository

- Contains Node.js app + DevOps

#### Option 2: Two-Repository (Recommended)

- Repo 1 → Node.js Retail App
- Repo 2 → DevOps Infrastructure

Pipeline flow:

```
Git Push → GitHub Actions → Docker Build → Docker Hub → EKS Deployment → Load Balancer → Users
```

Kubernetes performs **Zero-Downtime Rolling Updates**, ensuring continuous availability.

# 2. System Design

## 2.1 Application Design

### Architecture: Two-Tier

### Frontend Layer

- Can be:

- EJS / React / Static HTML
- Runs via Express server

# Backend Layer

- Node.js + Express API

### Example Structure

```
src/
 ├── routes/
 ├── controllers/
 ├── models/
 └── app.js
```

# Container Design

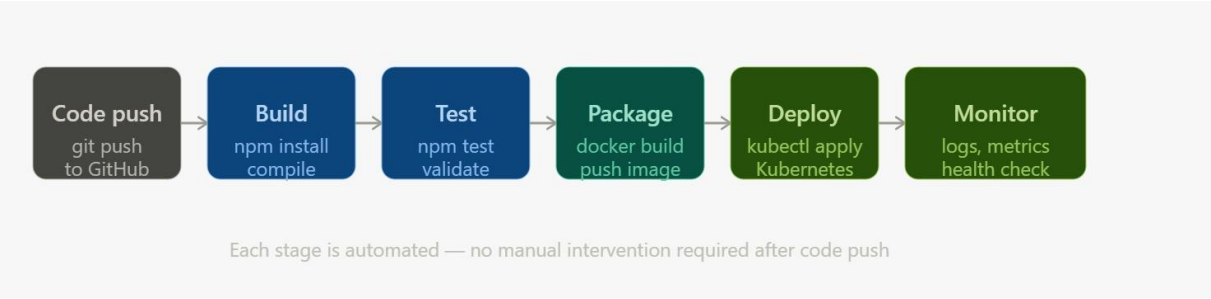
| Property | Value |
|---|---|
| Base Image | node:18-alpine |
| Port | 3000 |
| Entry | node server.js |

Steps:

- Copy package.json
- npm install
- Copy source
- Start server

### 2.2 Process Flow

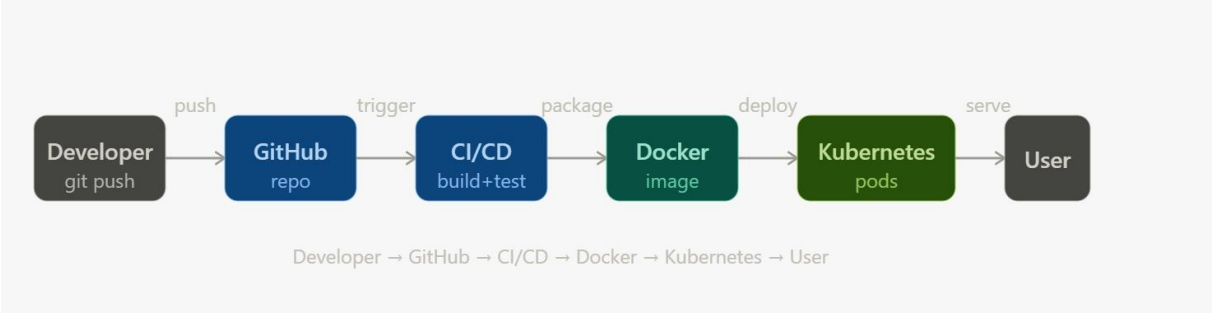
The end-to-end process flow from code change to live deployment is as follows:



## 2.3 Information Flow

The information flow describes how data moves through the system at runtime, from the user's browser to the AI agents and back.

**User Request Flow:**




## 2.4 Components Design

**Component 1- Dockerfile**

| Property | Value |
|---|---|
| Base Image | node:18-alpine |
| Port | 3000 |
| Command | node server.js |

Key steps: install system dependencies → copy requirements.txt → clone AI repo → install Python packages → create artifacts directory → set environment variables → define health check → set CMD.

**Component 2- Kubernetes Deployment (deployment.yaml)**

| Parameter | Value |
|---|---|
| replicas | 2 |
| strategy | RollingUpdate |
| maxSurge | 1 |
| maxUnavailable | 0 |

| | |
|---|---|
| containerPort | 3000 |

**Component 3- Kubernetes Service**

| Parameter | Value |
|---|---|
| type | LoadBalancer |
| port | 80 |
| targetPort | 3000 |

**Component 4- GitHub Actions Pipeline**

| Step | Action |
|---|---|
| 1 | Checkout |
| 2 | Docker Login |
| 3 | Build Image |
| 4 | Push Image |
| 5 | AWS Auth |
| 6 | Deploy |
| 7 | Verify |

**Component 5- AWS Infrastructure**

| Component | Config |
|---|---|
| EKS | ap-south-1 |
| Nodes | t3.small |
| Registry | Docker Hub |
| Load Balancer | AWS ELB |

## 2.5 Key Design Considerations

1.

   a **Zero Downtime Deployment**

2. `maxUnavailable: 0`
   a **Health Checks**
3. Node.js / endpoint
   a **Security**
4. Secrets stored in GitHub
   a **Scalability**
5. EKS auto-scaling
   a **Cost Efficiency**
6. t3.small instances

## 2.6 API Catalogue

Although this is a DevOps project with no custom REST API, the following endpoints are relevant to the system's operation:

**Internal Health Endpoints (Streamlit Built-in)**

| Endpoint | Method | Purpose |
|---|---|---|
| / | GET | Health check |

**Response:**

HTTP 200 OK

Body: "ok"

**External API Used by Application**

| API | Purpose |
|---|---|
| Payment API | Checkout |
| Product API | Inventory |

**Docker Hub Registry API (used by GitHub Actions)**

| Action | Command | Purpose |
|---|---|---|
| Push Image | docker push tiwariom/multi-agentmanufacturing:latest | Upload built image to registry |
| Pull Image | Handled by EKS automatically | EKS pulls image on pod creation |

**AWS APIs (used by GitHub Actions)**

| API | Command | Purpose |
|---|---|---|
| EKS kubeconfig | aws eks updatekubeconfig | Authenticate kubectl with EKS cluster |
| Kubernetes API | kubectl apply -f deployment.yaml | Create/update deployment on cluster |
| Kubernetes API | kubectl apply -f service.yaml | Create/update LoadBalancer service |
| Kubernetes API | kubectl rollout status | Verify deployment completed successfully |

## 3 Interfaces

- User Interface → Browser
- DevOps Interface → GitHub, AWS, Docker
- System Flow →

```
Browser → LoadBalancer → Service → Pod
```

## 4 State and Session Management

- Stateless Node.js containers
- Session handled via:
- JWT / cookies (optional)

.

## 5 Caching

The Dockerfile is structured to maximize Docker's layer cache. System dependencies and pip packages are installed before the git clone step so if only the AI repo code changes, Docker reuses all cached layers and skips straight to cloning. This reduces rebuild time significantly. imagePullPolicy: Always is set in the deployment manifest to ensure Kubernetes always pulls the latest image on every deploy, bypassing node-level cache. No application-level caching is implemented every agent run makes a fresh live call to the Gemini LLM API, which is intentional as supplier data must always be current.

## 6 Functional Requirements

### 6.1 User Authentication and Authorization

- Optional login system

### 6.2 Infrastructure Configuration Management

- Git-based version control

### 6.3 Provisioning Control and Monitoring

- Kubernetes probes

### 6.4 Dockerfile (Application Containerization)

- Uses Node.js base image
- Installs dependencies

### 6.5 GitHub Actions (Workflow Configuration)

- Auto trigger on push
- Fully automated deployment

### 6.6 CI/CD Strategy

- Every push triggers deployment
- Rolling update ensures no downtime

## 7 Non-Functional Requirements

- **Availability** → Always running (2 replicas)
- **Scalability** → Auto scaling
- **Security** → Secrets management
- **Performance** → Low latency (Mumbai region)
- **Maintainability** → Automated CI/CD
- **Cost Efficiency** → Low-cost EC2

## 8 References

- The following resources were referred to during the development and implementation of the project:

- • Official Node.js Documentation – https://nodejs.org
- • Docker Documentation – https://docs.docker.com
- • Kubernetes Documentation – https://kubernetes.io/docs
- • Git Documentation – https://git-scm.com/docs
- • GitHub Documentation – https://docs.github.com
- • GitHub Actions Documentation – https://docs.github.com/en/actions

## 9 GitHub

- https://github.com/ARSH2103/CI-CD-for-Node.js-Retail-App-.git